Fluids and Lubricants Specifications

MTU Fluids and Lubricants Specifications for Series 1600
Application C&I, Genset, Marine, Oil & Gas and Rail

A001063/03E

Power. Passion. Partnership.

© 2019 Copyright MTU Friedrichshafen GmbH

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of MTU Friedrichshafen GmbH. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. MTU Friedrichshafen GmbH reserves the right to change, delete or supplement the information provided as and when required.

Table of Contents

DCL-ID: 0000005579 - 005

1 Preface

1.1 General

Used symbols and means of representation

The following instructions are highlighted in the text and must be observed:

Important information
This field contains product information which is important or useful for the user. It refers to instructions, work and activities that have to be observed to prevent damage or destruction to the material.

Note:

A note provides special instructions that must be observed when performing a task.

Fluids and lubricants

The service life, operational reliability and function of the drive systems are largely dependent on the fluids and lubricants employed. The correct selection and treatment of these fluids and lubricants are therefore extremely important. This publication specifies which fluids and lubricants are to be used.

Test standard	Designation
DIN	Federal German Standards Institute
EN	European Standards
ISO	International Standards Organization
ASTM	American Society for Testing and Materials
IP	Institute of Petroleum
DVGW	German Gas and Water Industry Association

Table 1: Test standards for fluids and lubricants

Applicability of this publication

The Fluids and Lubricants Specifications will be amended or supplemented as necessary. Prior to use, ensure that the most recent version is available. The most recent version can be called up under:

http://www.mtu-online.com/mtu/technische-info/betriebsstoffvorschriften/index.de.html

If you have further queries, please contact your MTU representative.

Warranty

Use of the approved fluids and lubricants, either under the brand name or in accordance with the specifications given in this publication, constitutes part of the warranty conditions.

The supplier of the fluids and lubricants is responsible for the worldwide standard quality of the named products.

Important information
Fluids and lubricants for drive systems may be hazardous materials. Certain regulations must be obeyed when handling, storing and disposing of these substances.

These regulations are contained in the manufacturers' instructions, such as product-specific safety data sheets, statutory regulations and technical guidelines valid in the individual countries. Great differences can apply from country to country and a generally valid guide to applicable regulations for fluids and lubricants is therefore not possible within this publication.

TIM-ID: 0000018549 - 006

Users of the products named in these specifications are therefore obliged to inform themselves of the locally valid regulations. MTU accepts no responsibility whatsoever for improper or illegal use of the fluids and lubricants which it has approved.

Preservation

All information on preservation, representation and depreservation including the approved preservatives is available in the MTU Preservation and Represervation Specifications (publication number A001070/...). The most recent version can be called up under:

http://www.mtu-online.com/mtu/technische-info/konservierungs-und-nachkonservierungsvorschrift/index.de.html

TIM-ID: 0000018549 - 006

2 Engine Oils

2.1 Requirements and oil change intervals

Important
Dispose of used fluids and lubricants in accordance with local regulations. Used oil must never be disposed of via the fuel tank!

Requirements of the engine oils for MTU approval

The MTU conditions for engine oil approval are specified in the MTU Factory Standard MTL 5044, which can be ordered under this reference number.

Manufacturers of engine oils are notified in writing if their product is approved.

Diesel engine oils approved for Series 1600 engines are divided into the following MTU quality categories:
- Oil category 2: Higher quality oils / multi-grade oils
- Oil category 2.1: Multi-grade oils with a low ash-forming additive content (low SAPS oils)
- Oil category 3: Highest quality / Multi-grade oils
- Oil category 3.1: Multi-grade oils with a low ash-forming additive content (low SAPS oils)

Low SAPS oils are oils with a low sulfur and phosphor content and an ash-forming additive content of ≤ 1 %. They are only approved if the sulfur content in the fuel does not exceed 50 mg/kg. Depending on the exhaust aftertreatment used, the use of low-ash oils is prescribed (→ Page 37).

Selection of a suitable engine oil is based on fuel quality, projected oil drain interval and on-site climatic conditions. At present there is no international industrial standard which alone takes into account all these criteria.

Important
The use of engine oils not approved by MTU can result in increased wear and can mean that statutory emission limits can no longer be observed. This can be a punishable offense.

Special features for Rolls Royce Power Systems (RRPS)/MTU engine oils

The following multi-grade oils are available from MTU / MTU Detroit Diesel depending on the region.

Manufacturer & sales region	Product name	SAE grade	Oil category	Material no.
MTU Friedrichshafen Europe Middle East Africa	Diesel Engine Oil DEO SAE 15W-40	15W-40	2	20 l canister: X00070830 210 l barrel: X00070832 IBC: X00070833 Loose items: X00070835 (only on request)
MTU America Americas	Power Guard® SAE 15W-40 Off Highway Heavy Duty	15W-40	2.1	5 gallons: 800133 55 gallons: 800134 IBC: 800135
MTU Asia Asia	Diesel Engine Oil DEO SAE 15-W40	15W-40	2	18 l canister: 64247/P 200 l barrel: 65151/D
MTU Asia China	Diesel Engine Oil - DEO 15W-40	15W-40	2	20 l canister: 64242/P 205 l barrel: 65151/D
	Diesel Engine Oil - DEO 10W-40	10W-40	2	20 l canister: 60606/P
	Diesel Engine Oil - DEO 5W-30	5W-30	3	20 l canister: 60808/P

TIM-ID: 0000010717 - 004

Manufacturer & sales region	Product name	SAE grade	Oil category	Material no.
MTU Asia Indonesia	Diesel Engine Oil - DEO 15W-40	15W-40	2	20 l canister: 64242/P 205 l barrel: 65151/D
MTU India Pvt. Ltd. India	Diesel Engine Oil - DEO 15W-40	15W-40	2	20 l canister: 63333/P 205 l barrel: 65151/P

Table 2: Multi-grade oils from RRPS/MTU

Oil change interval

Important

The oil change interval is 1,000 operating hours or max. 1 year under the condition that engine oils of oil category 3 and 3.1 and approved fuels (→ Page 17) are used.
The oil change interval is 500 operating hours or max. 1 year under the condition that engine oils of oil category 2 and 2.1 and approved fuels (→ Page 17) are used.
If fuels which have not been approved are used, shorter oil change intervals are to be expected.
Prior to using unapproved fuels, contact MTU Friedrichshafen GmbH to determine the applicable oil change intervals.

Important

Mixing different engine oils is strictly prohibited!
If, in exceptional cases, the engine oil filled in the engine is not available, top up with another mineral or synthetic engine oil. Ensure that it is approved for MTU products (→ Page 37).
Note the following:

- If you top up an engine oil of lower quality, the maintenance interval corresponding to the lower quality (oil category) must apply. The maintenance interval is reduced.
- If you use an engine with a higher quality, the maintenance interval remains as it is. Observe the specifications in the maintenance booklet.

Changing to another oil grade can be done together with an oil change. The remaining oil quantity in the engine oil system is not critical in this regard. This procedure also applies to MTU's own engine oil in the regions Europe, Middle East, Africa, America and Asia.

Important

When changing to an engine oil of Category 3, note that the improved cleaning effect of these engine oils can result in the loosening of engine contaminants (e.g. carbon deposits).
It may be necessary therefore to reduce the oil change interval and oil filter service life (one time during change).

TIM-ID: 0000010717 - 004

2.2 Viscosity grades

Selection of the viscosity grade is based primarily on the ambient temperature at which the engine is to be started and operated. Figure (→ Figure 1) contains guideline values for the temperature limits of the individual viscosity grades.

The temperature specifications of the SAE grade are always based on fresh oils. During operation, engine oil ages due to soot and fuel residue. This results in significant deterioration of the properties of the engine oil particularly at low outside temperatures. At outside temperatures below -20 °C, MTU strongly recommends the use of engine oil of SAE grade 5W-30 or - if approved - 0W-30.

If the prevailing temperature is too low, the engine oil must be preheated.

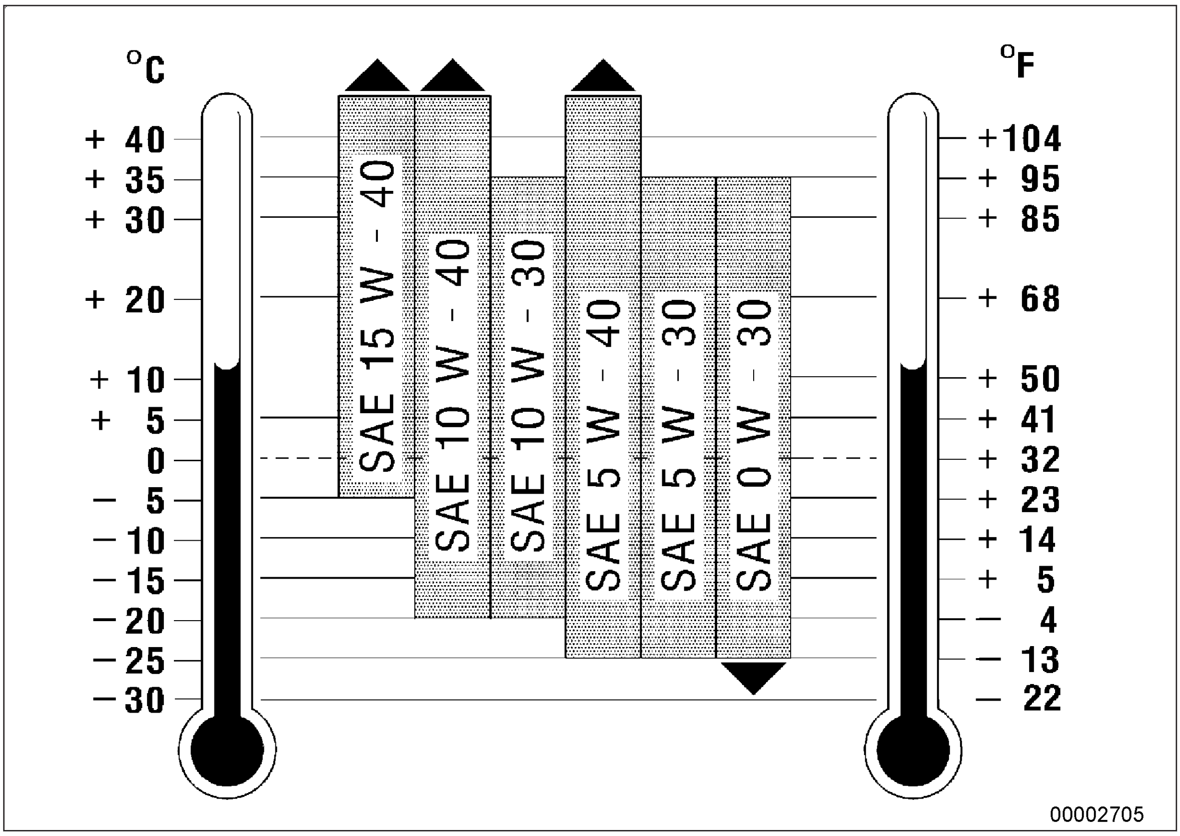

Figure 1: Viscosity grades

TIM-ID: 0000034716 - 002

2.3 Used-oil analysis

In order to check the used oil, it is recommended that regular oil analyses be carried out. Oil samples should be taken and analyzed at least once per year and during each oil change and under certain conditions, depending on application and the engine's operating conditions, sampling / analysis should take place more frequently.

The specified test methods and limit values (Analytical Limit Values for Used Diesel Engine Oils) (→ Table 3) indicate when the results of an individual oil sample analysis are to be regarded as abnormal.

An abnormal result requires immediate investigation and remedy of the abnormality.

The limit values relate to individual oil samples. When these limit values are reached or exceeded, an immediate oil change is necessary. The results of the oil analysis do not necessarily give an indication of the wear status of particular components.

In addition to the analytical limit values, the engine condition, its operating condition and any operational faults are decisive factors with regard to oil changes.

Some of the signs of oil deterioration are:
- Abnormally heavy deposits or precipitates in the engine or engine-mounted parts such as oil filters, centrifugal oil filters or separators, especially in comparison with the previous analysis.
- Abnormal discoloration of components.

Analytical limit values for used diesel engine oils

	Test Method	Limit values	
Viscosity at 100 °C max. mm²/s	ASTM D445 DIN 51562	SAE 5W-30 SAE 10W-30	15.0
		SAE 5W-40 SAE 10W-40 SAE 15W-40 SAE 20W-40	19.0
min. mm²/s		SAE 5W-30 SAE 10W-30	9.0
		SAE 5W-40 SAE 10W-40 SAE 15W-40 SAE 20W-40	10.5
Flash point °C (COC)	ASTM D92 DIN EN ISO 2592	Min. 190	
Flash point °C (PM)	ASTM D93 ISO 2719	Min. 140	
Soot content (by weight %)	DIN 51452 CEC-L-82-A-97	Max. 3,5	
Total base number (mg KOH/g)	ASTM D2896 ISO 3771 DIN 51639	Min. 50% of new-oil value	
Proportion of water (vol. %)	ASTM D6304 EN 12937 ISO 6296	Max. 0.2	
Oxidation (A/cm)[1]	DIN 51453[1]	Max. 25	
Ethylene glycol (mg/kg)	ASTM D2982	Max. 100	

Table 3:

[1] = only possible if there are no ester compounds

TIM-ID: 0000083718 - 001

3 Coolants

3.1 Coolant – General

Coolant definition

Coolant	= coolant additive (concentrate) + fresh water to predefined mixing ratio Ready for use in engine

Coolants must be prepared from suitable fresh water and a coolant additive which has been approved by MTU Friedrichshafen GmbH.

Important

Conditioning of the coolant takes place outside the engine.
Mixing of different coolant additives and supplementary additives is prohibited!

Important

Ready mixtures are coolants ready for direct use in the engine. They must not be diluted with fresh water.

Important

Flush with fresh water before changing from an antifreeze product containing silicate (ready mix or concentrate) to a silicate-free product! The same applies when changing from a silicon-free corrosion-inhibiting antifreeze to a product containing silicon.

The approval conditions for coolant additives are defined in the MTU delivery standard MTL 5048 / corrosion-inhibiting antifreezes.

Emulsifiable corrosion inhibitor oils and water-soluble corrosion inhibitors are not permitted for Series 1600.

Coolant manufacturers are informed in writing if their product is approved by MTU.

Permissible application concentrations of engine coolants

Concentration for use	Coolant additive	Fresh water	Antifreeze protection[1] up to approx.
Minimum	40% by volume	60% by volume	-27 °C
	50% by volume	50% by volume	-37 °C
Maximum	55% by volume	45% by volume	-45 °C

Table 4: Coolant mixing ratio and limit values

[1] = antifreeze specifications determined as per ASTM D 1177

The coolant concentration used mainly depends on the antifreeze protection requirements.

Note:

The application concentration of a coolant or a ready mixture must be specified such that the share of coolant additives is always named first.

Example:

Coolant concentration 40% by volume = 40% by volume coolant additive + 60% by volume Fresh water

For ready mixtures, the proportion of coolant additive (concentrate) is always named first.

Example:

Power Cool®Universal 50/50 mix = 50% by volume coolant additive / 50% by volume Fresh water

TIM-ID: 0000035549 - 004